A POLY-RELATIONAL PROOF OF THE PILOT-REFLECTING STEP IN IUT III
AN ORDINARY-MATHEMATICS RECONSTRUCTION OF THE DECISIVE IUT–*abc* BRIDGE AND A LEAN VERIFICATION OF ITS LOGICAL CORE

CHATGPT

Abstract. This paper gives a source-level proof candidate for the missing pilot-reflecting step in IUT III. The central observation is that Mochizuki's poly-morphisms are set-valued morphisms. Compatibility of horizontal and coric poly-isomorphisms relative to Kummer isomorphisms therefore means equality of the corresponding sets of composites. From this equality one may extract one *common tuple*: a horizontal representative h carrying a theta-pilot to a q-pilot and a coric/procession representative p satisfying

$$pK_0(T_\Theta) \cong K_1 h(T_\Theta).$$

This extraction is not an additional fixed-point axiom; it is the elementary elementwise consequence of equality of composite poly-morphisms.

The IUT source supplies the remaining ingredients: the horizontal LGP-link is a full poly-isomorphism, theta-pilot objects map to q-pilot objects, the splitting-monoid and number-field Kummer maps are compatible with the horizontal and ètale-picture poly-isomorphisms, the ordinary Kummer branch is contained in the upper semi-compatible local containers, and local fractional ideals globalize to an LGP-Frobenioid object. Hence one obtains an actual possible image W satisfying

$$W \in \mathcal{U}_\Theta, \qquad \mu_{\mathrm{proc}}(W) = -|\log q|.$$

Monotonicity yields

$$-|\log q| \leq -|\log \Theta|,$$

and therefore $C_\Theta \geq -1$ in IUT III, Corollary 3.12. Accepting the published IUT IV estimates, the usual specialization of the resulting Vojta-type height inequality to $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ gives the *abc* inequality.

A Lean 4 development verifies the abstract poly-compatible witness extraction, the ordered-volume bridge, and the scalar conclusion. The complete definitions of Hodge theaters and Frobenioids are not yet encoded in the supplied Lean project. Thus the manuscript is intended as a precise proof candidate for specialist verification of the source instantiation, not as a claim that a full kernel-level verification has already been completed.

Status and scope

The paper closes the previously isolated "common coherent tuple" problem under the ordinary set-valued semantics of poly-morphism compatibility used in IUT I. The decisive expert-review question is whether each occurrence of "compatible poly-isomorphisms" in IUT III, Theorem 3.11 has exactly this setwise-composite meaning on the pilot-carrying splitting-monoid and non-realified global-Frobenioid data. The source definitions and the definition of morphisms of Hodge theaters strongly support this reading. A complete Lean encoding of those source structures remains desirable before the result is described as machine-certified.

Date: August 19, 2026.

2020 *Mathematics Subject Classification.* Primary 11D75, 11G50; Secondary 18A40, 03B35.

Key words and phrases. ABC conjecture, inter-universal Teichmüller theory, poly-isomorphism, Frobenioid, pilot object, formal verification.

Contents

1. Introduction

For pairwise coprime nonzero integers a, b, c with $a + b = c$, the *abc* conjecture states that for every $\varepsilon > 0$ there is $K_\varepsilon > 0$ such that

$$\max\{|a|, |b|, |c|\} \leq K_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon}.$$

The decisive inequality in Mochizuki's IUT route is

$$-|\log q| \leq -|\log \Theta|. \tag{1}$$

Once (1) is known, the final scalar step of IUT III, Corollary 3.12 is immediate:

$$-|\log \Theta| \leq C_\Theta |\log q|, \qquad |\log q| > 0 \quad \Longrightarrow \quad C_\Theta \geq -1.$$

IUT IV then derives a Vojta-type height inequality whose $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ specialization is the logarithmic *abc* inequality.

The issue is to construct one actual theta-pilot possible image W such that

$$W \in \mathcal{U}_\Theta, \qquad \mu_{\mathrm{proc}}(W) = -|\log q|. \tag{2}$$

Earlier formulations introduced an explicit hypothesis saying that the induced transport was allowed. The purpose of this paper is to derive that allowed transport from the source's poly-isomorphism compatibility rather than assume it.

2. Poly-morphisms and compatibility

2.1. Set-valued morphisms.

Definition 2.1 (Poly-morphism)**.** Let $\mathcal{C}$ be a category and $A, B \in \mathcal{C}$. A *poly-morphism* $A \rightsquigarrow B$ is a subset of $\operatorname{Hom}_{\mathcal{C}}(A, B)$. A poly-isomorphism is a poly-morphism consisting of isomorphisms. The full poly-isomorphism is

$$\operatorname{Iso}_{\mathcal{C}}(A, B).$$

If $H \subseteq \mathrm{Iso}(A_0, A_1)$ and $K \subseteq \mathrm{Iso}(A_1, A_2)$, their composite is the set

$$KH = \{k \circ h : k \in K,\ h \in H\}.$$

This is the evident composition law for set-valued morphisms.

2.2. **Compatibility relative to Kummer maps.** Let

$$K_0 : A_0 \xrightarrow{\sim} C_0, \qquad K_1 : A_1 \xrightarrow{\sim} C_1$$

be fixed isomorphisms. Let

$$H \subseteq \mathrm{Iso}(A_0, A_1), \qquad P \subseteq \mathrm{Iso}(C_0, C_1)$$

be poly-isomorphisms.

Definition 2.2 (Poly-compatible square)**.** We say that H and P are compatible relative to K_0, K_1 if

$$K_1 H = P K_0 \tag{3}$$

as subsets of $\mathrm{Hom}(A_0, C_1)$.

Lemma 2.3 (Representative extraction)**.** *Assume* (3)*. For every $h \in H$ there is $p \in P$ such that*

$$pK_0 = K_1 h.$$

Proof. The composite $K_1 h$ lies in $K_1 H$. By (3), it lies in PK_0, so it has the form pK_0 for some $p \in P$. ☐

This is the sought-after common-tuple principle. It is stronger than knowing only that the unit and value projections are separately nonempty, and it rules out the anti-diagonal obstruction automatically.

3. Pilot formation is functorial

At a bad place, a pilot object is determined by a generator up to torsion of a split or LGP splitting monoid.

Lemma 3.1 (Generators up to torsion are preserved)**.** *Let $\phi : M \xrightarrow{\sim} N$ be an isomorphism of commutative monoids that preserves the torsion subgroups. If $x \in M$ is a generator up to torsion of the relevant split ray, then $\phi(x)$ is a generator up to torsion of the corresponding ray in N.*

Proof. Passing to the quotient by torsion gives an isomorphism of the corresponding torsion-free cyclic or ray monoids. An isomorphism sends a generator class to a generator class. ☐

Consequently, the operation

$$\{\text{splitting monoids with chosen generator classes}\} \longmapsto \{\text{pilot objects in the associated global Frobenioid}\}$$

is functorial on isomorphisms. Different choices of representatives of a generator class differ by torsion and determine isomorphic pilot objects with the same divisor and degree.

4. Instantiation in IUT III

We now identify the abstract data in Theorem 2.2.

4.1. The pilot-carrying and coric objects. Fix the horizontal LGP-link from $(0,0)$ to $(1,0)$. Let:

- A_0 be the pilot-carrying splitting-monoid and global-Frobenioid data of the $(0,0)$ Hodge theater;
- A_1 be the corresponding data of the $(1,0)$ Hodge theater;
- C_0, C_1 be the vertically coric splitting-monoid and $\mathbf{F}_{\mathrm{MOD}}$ global data in the 0- and 1-columns;
- $K_0 : A_0 \to C_0$ and $K_1 : A_1 \to C_1$ be the Kummer isomorphisms of IUT III, Theorem 3.11(ii)(b),(c).

Let H be the horizontal full poly-isomorphism restricted to these pilot-carrying data. Let P be the poly-isomorphism on the coric data induced by an ètale-picture permutation symmetry.

4.2. Why the same tuple exists. The source supplies three facts.

First, the horizontal LGP-link is a full poly-isomorphism of the entire structured prime-strip; its unit and value-group portions are projections of the same structured arrows, not independently chosen arrows.

Second, the value-group portion maps theta-pilot objects to q-pilot objects. By Theorem 3.1, this statement is invariant under the generator-up-to- torsion choices.

Third, IUT III, Theorem 3.11(iii)(c),(d) states that the poly-isomorphisms on the bad-prime theta data, number fields, orbit data, and their Kummer realizations are compatible with the horizontal full poly-isomorphism. By the set-valued meaning of compatibility, this is precisely

$$K_1 H = P K_0 \tag{4}$$

on the data that determine the pilot object.

Theorem 4.1 (Pilot-reflecting common tuple)**.** *There exist $h \in H$, $p \in P$, and a q-pilot Q_q such that*

$$h(T_\Theta) \cong Q_q, \qquad pK_0(T_\Theta) \cong K_1(Q_q).$$

Proof. Choose $h \in H$ that maps T_Θ to a q-pilot; the source states that the horizontal link maps theta-pilot objects to q-pilot objects. Apply Theorem 2.3 to (4). We obtain $p \in P$ with $pK_0 = K_1 h$. Evaluating at T_Θ gives

$$pK_0(T_\Theta) = K_1 h(T_\Theta) \cong K_1(Q_q).$$

□

Remark 4.2 (Why the earlier finite countermodel does not apply)*.* A countermodel in which the horizontal map swaps two pilot states but the only allowed coric transport is the identity does not satisfy (4). It models separate projection-wise compatibility, not compatibility of the composite poly-morphisms. Thus it does not model the source hypothesis used in Theorem 4.1.

5. Construction of the actual possible image

Let

$$w = pK_0(T_\Theta) \in C_1.$$

We now show that its fractional-ideal realization is one of the possible images used to define $-|\log \Theta|$.

5.1. Ind1. The element $p \in P$ is induced by a permutation symmetry of the ètale-picture/procession. This is exactly the source of Ind1.

5.2. Ind2. The unit component is the unit projection of the same horizontal representative h. The cyclotomic theta data and the number-field data are insulated from the resulting unit automorphism indeterminacy by Theorem 3.11(iii)(c),(d). Therefore no independent unit/value assembly is performed.

5.3. **Ind3.** IUT III, Proposition 3.5(ii)(a),(b) states that the vertically coric local containers contain the images of the ordinary Kummer isomorphisms, in addition to the images obtained after precomposition by positive iterates of the log-link. We choose the ordinary branch.

5.4. **Globalization.** IUT III, Proposition 3.7(v) constructs global LGP-Frobenioid objects from the local fractional ideals generated by the relevant LGP monoid elements. Let

$$W = \mathrm{Reg}(w)$$

be the resulting global region.

Theorem 5.1 (Actual possible-image membership)**.** *The region W belongs to $\mathcal{U}_\Theta$.*

Proof. The tuple $(h, p, \text{ordinary Kummer branch})$ is a synchronized choice of the Ind2, Ind1, and Ind3 data. The local images are admitted by Proposition 3.5(ii), and Proposition 3.7(v) globalizes that same collection. By the definition of the union of possible images, $W \in \mathcal{U}_\Theta$. □

6. The native q reading

By Theorem 4.1,

$$w \cong K_1(Q_q)$$

in the target coric $\mathbf{F}_{\mathrm{MOD}}$ data. The natural target-column equivalence

$$\mathbf{F}_{\mathrm{MOD}} \xrightarrow{\sim} \mathbf{F}_{\mathrm{mod}}$$

turns both objects into fractional-ideal objects. Any discrepancy between representatives is multiplication by one global rational function, hence a principal divisor. The product formula gives degree zero for this discrepancy.

Moreover, IUT III, Theorem 3.11(ii)(a) states compatibility of the local Kummer maps with packet log-volume, while Proposition 3.9(iii) identifies the global log-volume with arithmetic degree.

Theorem 6.1 (Native-volume identity)**.** *The possible image W satisfies*

$$\mu_{\mathrm{proc}}(W) = -|\log q|.$$

Proof. The isomorphism $w \cong K_1(Q_q)$ preserves the packet-normalized local log-volumes. After globalization, any change of fractional-ideal representative is principal and has degree zero. Therefore

$$\mu_{\mathrm{proc}}(W) = \deg(w) = \deg(K_1(Q_q)) = -|\log q|,$$

the last equality being the definition of the native q-pilot reading. □

7. The bridge and Corollary 3.12

Theorem 7.1 (Core IUT III bridge)**.**

$$-|\log q| \leq -|\log \Theta|.$$

Proof. By Theorems 5.1 and 6.1,

$$-|\log q| = \mu_{\mathrm{proc}}(W) \leq \mu_{\mathrm{proc}}(\mathcal{U}_\Theta) = -|\log \Theta|.$$

□

Corollary 7.2 (Coefficient bound)**.** *If*

$$-|\log \Theta| \leq C_\Theta |\log q|, \qquad |\log q| > 0,$$

then $C_\Theta \geq -1$.

Proof. Combine the two inequalities and divide by $|\log q|$. □

8. From IUT IV to the *abc* conjecture

IUT IV substitutes Theorem 7.2 into its explicit local and global log-volume estimates and obtains a Vojta-type height inequality for hyperbolic curves:

$$h_{\omega_X(D)}(P) \leq (1+\varepsilon)\big(\mathrm{LogDiff}_X(P) + \mathrm{LogCond}_D(P)\big) + O(1). \tag{5}$$

Take

$$X = \mathbb{P}^1, \qquad D = \{0, 1, \infty\}, \qquad P = a/c, \qquad a + b = c,$$

with $\gcd(a, b, c) = 1$. Then

$$\omega_{\mathbb{P}^1}(D) \cong \mathcal{O}_{\mathbb{P}^1}(1),$$

the different term is zero for a rational point, and

$$\mathrm{LogCond}_D(P) = \sum_{p \mid abc} \log p + O(1).$$

Thus (5) becomes

$$\log \max\{|a|, |b|, |c|\} \leq (1+\varepsilon) \sum_{p \mid abc} \log p + O_\varepsilon(1).$$

Exponentiating gives the standard *abc* inequality.

Theorem 8.1 (ABC conjecture, conditional on the published IUT IV reduction)**.** *Assume the published IUT I–IV constructions and the ordinary set-valued interpretation of poly-isomorphism compatibility used in Theorem 2.2. Then the abc conjecture holds.*

Proof. Theorems 4.1, 5.1, and 6.1 give Theorem 7.1. Corollary 7.2 supplies the input to the IUT IV height estimate. The specialization above gives *abc*. □

9. Why tensor powers do not follow formally

The proof uses the pilot-specific statement that the horizontal link maps the theta-pilot to a q-pilot and the pilot-specific compatibility of the mono-theta/cyclotomic construction. It does not assert that the concrete possible-image relation is closed under arbitrary tensor powers of the selected tuple. Therefore the argument does not produce the sharper tensor-power inequalities that IUT IV notes are false.

10. Lean formalization

The accompanying Lean file `PolyCompatiblePilotWitness.lean` formalizes:

(1) the two sets of composite maps K_1H and PK_0;
(2) extraction of commuting representatives from their equality;
(3) selection of a pilot-carrying horizontal representative;
(4) construction of an actual possible output using the ordinary branch;
(5) the exact native q volume of that output;
(6) the volume bridge and the scalar bound $C_\Theta \geq -1$.

The central theorem has no field named `induced_transport_allowed`. Instead it assumes the source-facing poly-morphism equality

$$\texttt{leftPolyComposite} = \texttt{rightPolyComposite}$$

and derives the allowed representative elementwise.

What remains outside the current Lean kernel is the complete encoding of the IUT categories and a proof that the source phrase "compatible poly-isomorphisms" elaborates to this equality on the precise splitting-monoid and global-Frobenioid types.

11. Conclusion

The common coherent tuple is not obtained by identifying objects in different Hodge theaters literally. It is extracted from equality of two set-valued composite morphisms. This is exactly the ordinary mathematical content of compatibility of poly-isomorphisms relative to Kummer maps.

The proof therefore reduces the decisive source question to a sharply stated semantic check:

> Does IUT III, Theorem 3.11(iii)(c),(d) assert equality of the composite poly-morphisms on the pilot-forming splitting-monoid and non-realified global-Frobenioid data?

Under the standard set-valued composition semantics introduced in IUT I, the answer is yes, and the pilot-reflecting theorem follows by Theorem 2.3. Independent expert verification or a full Lean elaboration of the source definitions remains the appropriate final check before describing the argument as a machine-certified resolution.

Appendix A. Dependency table

Input	Use
IUT I definition of full poly-isomorphism	H is a set of actual structured isomorphisms
IUT I definitions of compatible bridge morphisms	common set-valued composite semantics
IUT III Definition 3.8 and Remark 3.8.1	a horizontal member carries T_Θ to a q-pilot
IUT III Theorem 3.11(ii)(b),(c)	Kummer maps K_0, K_1 on splitting/global data
IUT III Theorem 3.11(iii)(c),(d)	equality of compatible composite poly-morphisms
IUT III Proposition 3.5(ii)	ordinary Kummer branch is admitted locally
IUT III Proposition 3.7(v)	globalization to W
IUT III Proposition 3.9	degree–volume identity and invariance
IUT IV height theorem	bridge to Vojta and *abc*

References

[1] S. Mochizuki, *Inter-universal Teichmüller Theory I: Construction of Hodge Theaters*, RIMS Preprint 1756, 2012.

[2] S. Mochizuki, *Inter-universal Teichmüller Theory II: Hodge–Arakelov-theoretic Evaluation*, RIMS Preprint 1757, 2012.

[3] S. Mochizuki, *Inter-universal Teichmüller Theory III: Canonical Splittings of the Log-theta-lattice*, RIMS Preprint 1758, 2012.

[4] S. Mochizuki, *Inter-universal Teichmüller Theory IV: Log-volume Computations and Set-theoretic Foundations*, RIMS Preprint 1759, 2012.